CSI Daily Monitor

Performance Monitoring & Forecasting System

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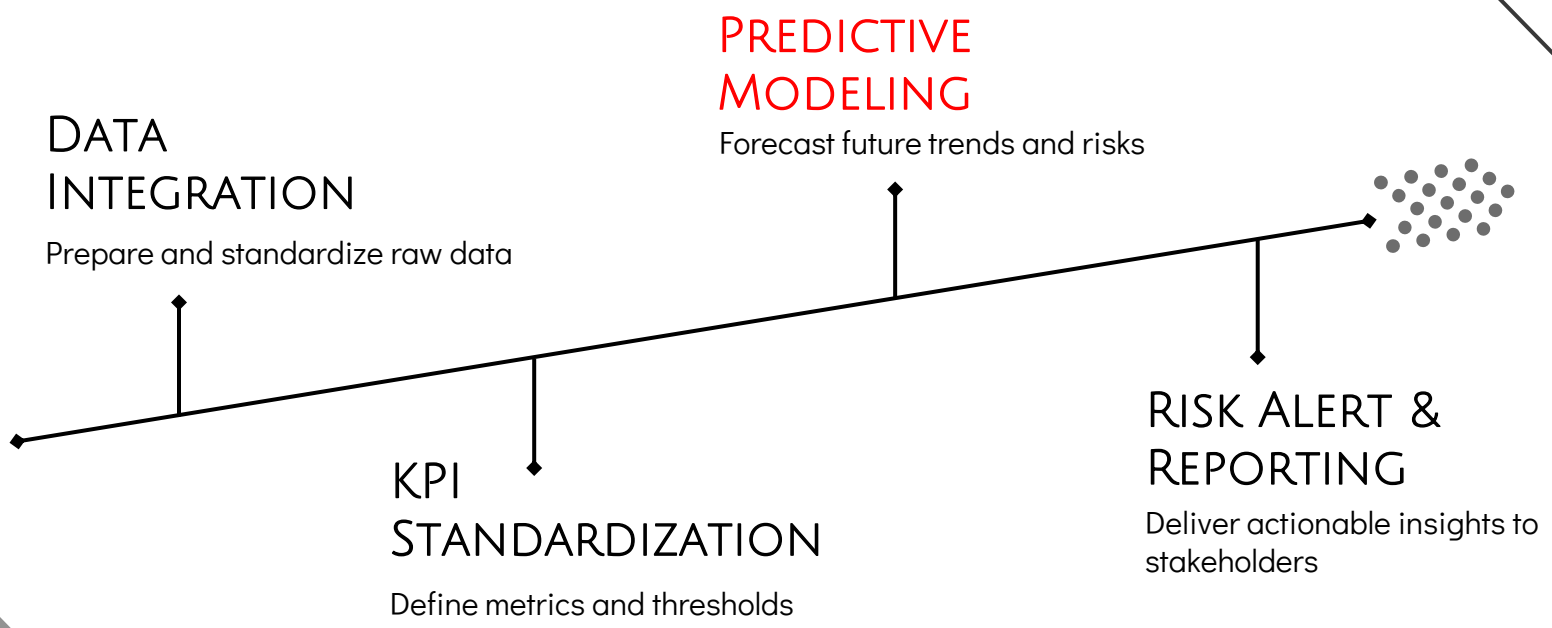
Environment : RStudio ChatGPT

The Challenge

“**Customer satisfaction is monitored too late**. Issues are only detected after they happen, not before they impact operations.”

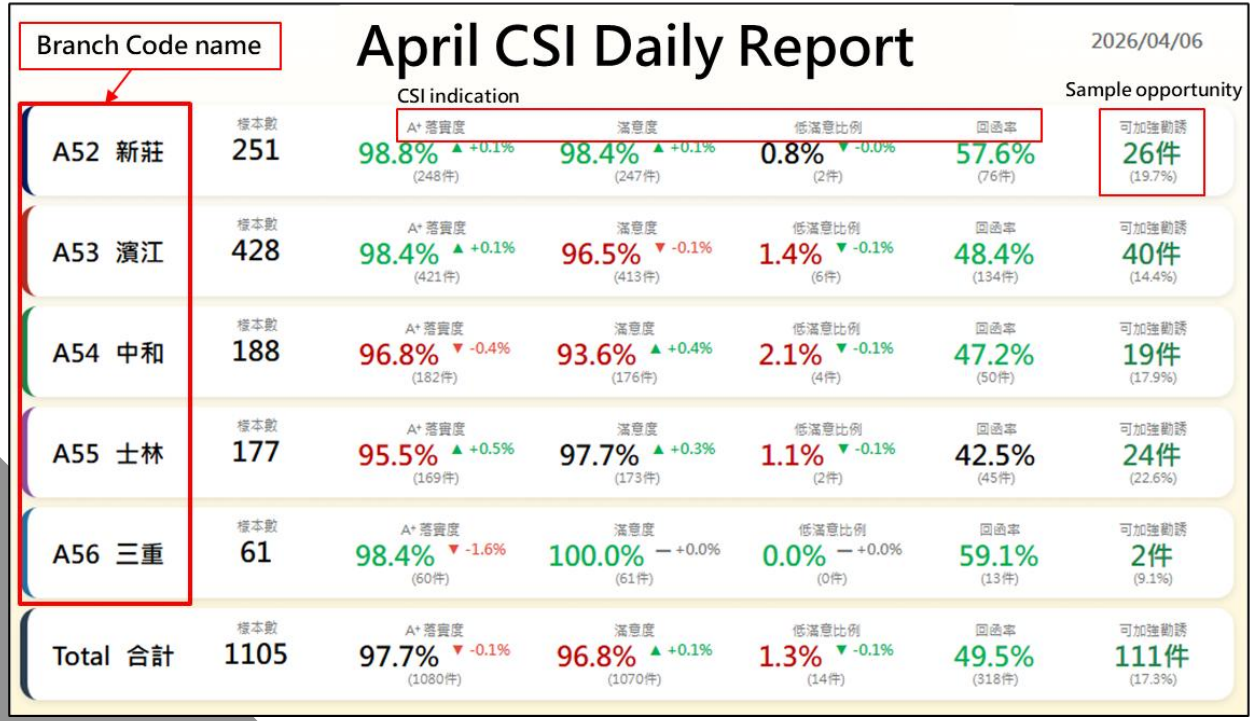
Objective

Shift from reactive reporting to proactive risk detection.



KPI Dashboard

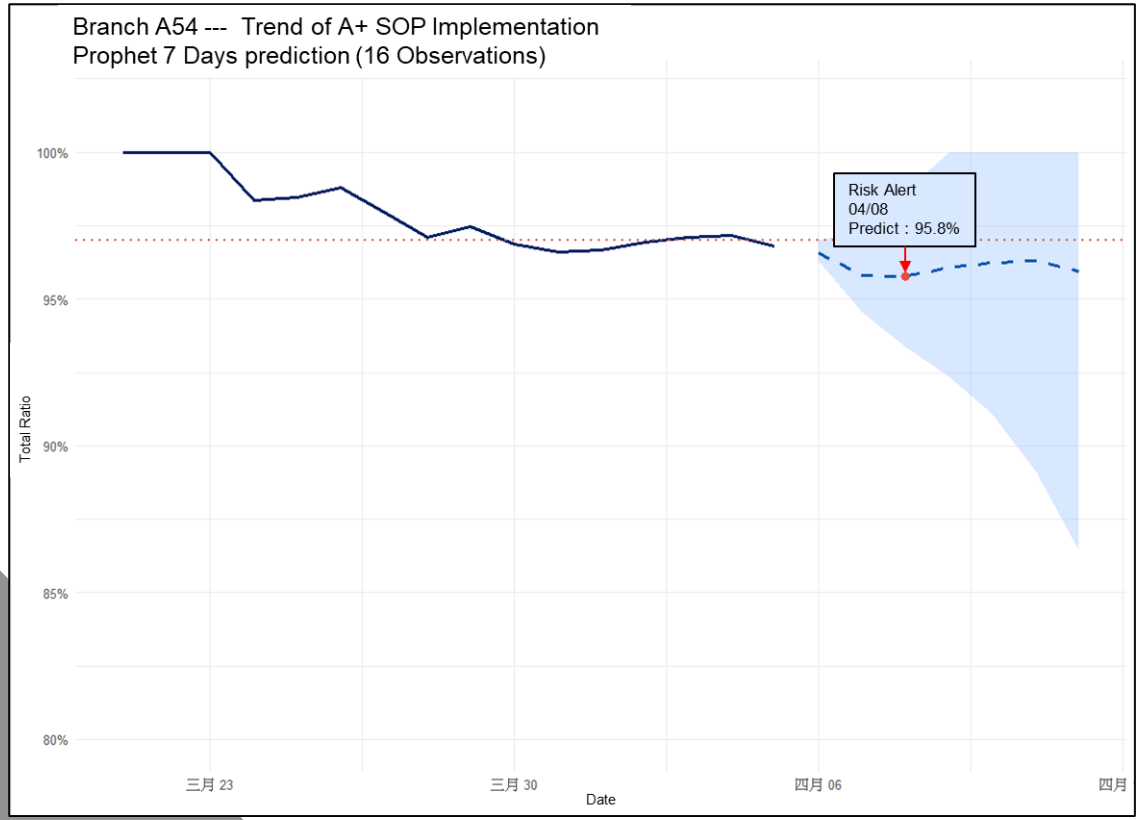
Driving Daily Operational Decisions



Provides real-time visibility across all branches
Highlights underperforming KPIs using color signals
Identifies actionable opportunities
Enables supervisors to take immediate action

Forecasting

Identifying Risks Before Happen



Using time-series models (ARIMA & Prophet) to forecast CSI trendsand identify potential performance risks in advance.

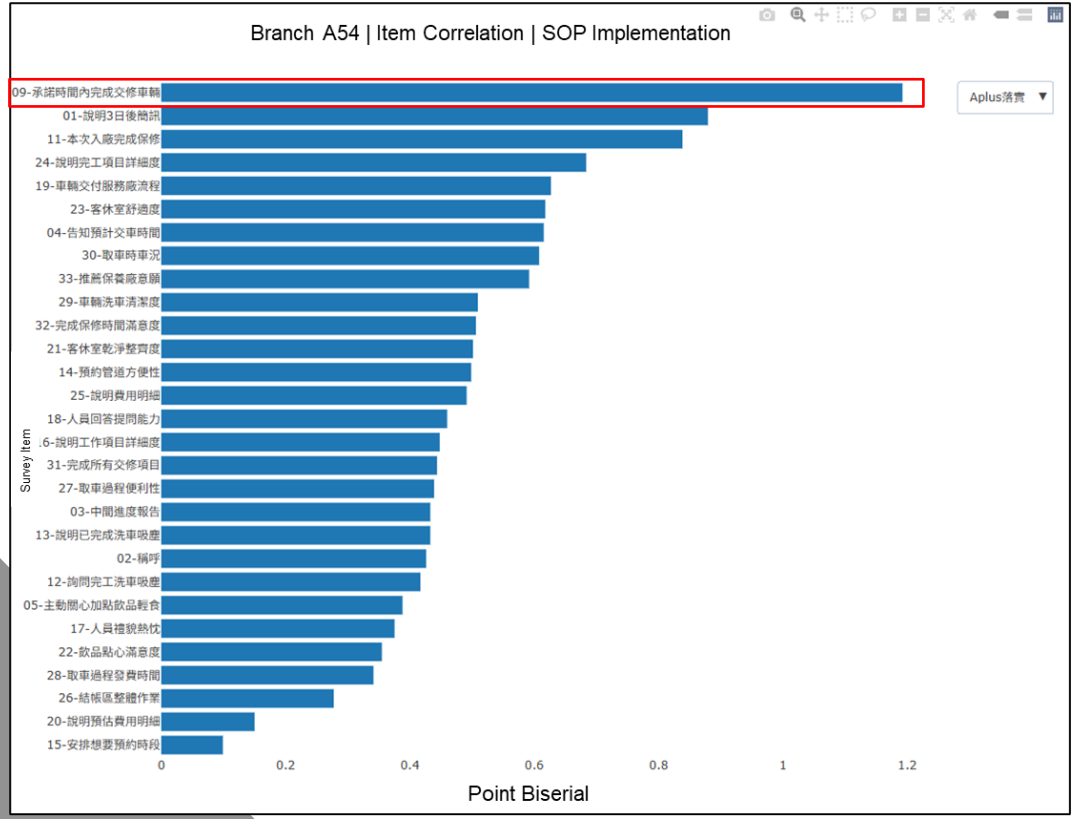
Key capabilities：

- Identifies the lowest prediction as the highest risk day.
- Provides early warning before KPI thresholds are breached.
- Supports proactive intervention instead of reactive reporting.

Predicted Branch A54 CSI drops to 95.8% on Apr 8, indicating a high risk of falling below the 97% target.

Action

What Drives Dissatisfaction & Where to Act



Using Point-Biserial Correlation method for precision key factors finding.

Key Findings：

- Multiple service factors strongly correlate with dissatisfaction
- Process-related issues have the highest negative impact
- These factors is the most critical of poor customer experience

Insight：

The most critical driver of dissatisfaction is“09-Service completion within expected time",showing the strongest negative correlation.Addressing this factor is expected to significantly reduce dissatisfaction.

Recommended Actions：

- Improve communication of waiting time during service
- Standardize service process to reduce variability across branches
- Prioritize training on high-impact service touchpoints

Business Impact —
From Insights to Measurable Outcomes

Projected Impact

- CSI improvement: +2% to +3%
- Reduction in low satisfaction cases: 15%–20%
- Earlier intervention window: 3–7 days
- Increased response efficiency across branches
- More focused allocation on high-impact factors

Business Value

- Enables proactive decision-making
- Reduces operational risks before KPI degradation
- Improves customer experience consistency across branches
- Transforms data into actionable operational strategy

This solution demonstrates how data can be translated intoproactive decision-making and measurable business impact.

THANK YOU

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